

CSE370 : Database Systems

Assignment 01 | Spring 2025

ID : 22201344 | Name : TAHSIN RAIHAN ROBBANI

No 1 Query (as Plain Text)	<pre>CREATE DATABASE Google_22201344; USE Google_22201344; CREATE TABLE Developers (member_id char(2), name varchar(30), email varchar(30), influence_count int, Joining_date date, multiplier int); SHOW TABLES; DESCRIBE Developers; INSERT INTO Developers values ('1', 'Taylor Otwell', 'otwell@laravel.com', '739360', '2020-6-10', 10), ('2', 'Ryan Dahl', 'ryan@nodejs.org', '633632', '2020-04-22', 10), ('3', 'Brendan Eich', 'eich@javascript.com', '939570', '2020-05-07', 8), ('4', 'Evan You', 'you@vuejs.org', '982630', '2020-06-11', 7), ('5', 'Rasmus Lerdorf', 'lerdorf@php.net', '937927', '2020-06-3', 8), ('6', 'Guido van Rossum', 'guido@python.org', '968827', '2020-07-18', 19), ('7', 'Adrian Holovaty', 'adrian@django project.com', '570724', '2020-05-07', 5), ('8', 'Simon Willison', 'simon@django project.com', '864615', '2020-04-30', 4), ('9', 'James Gosling', 'james@java.com', '719491', '2020-05-18', 5), ('10', 'Rod Johnson', 'rod@spring.io', '601744', '2020-05-18', 7), ('11', 'Satoshi Nakamoto', 'nakamoto@blockchain.com', '630488', '2020-05-10', 10);</pre>
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	<pre>SELECT * FROM Developers;</pre>
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No 1 SS
(of Query & Output
in Shell)

```
MariaDB [Google_22201344]> DESCRIBE Developers;
```

Field	Type	Null	Key	Default	Extra
member_id	char(2)	YES		NULL	
name	varchar(30)	YES		NULL	
email	varchar(30)	YES		NULL	
influence_count	int(11)	YES		NULL	
Joining_date	date	YES		NULL	
multiplier	int(11)	YES		NULL	

```
6 rows in set (0.011 sec)
```

```
MariaDB [Google_22201344]> INSERT INTO Developers values
```

```
-> ('1', 'Taylor Otwell', 'otwell@laravel.com', '739360', '2020-06-10', 10),
-> ('2', 'Ryan Dahl', 'ryan@nodejs.org', '633632', '2020-04-22', 10),
-> ('3', 'Brendan Eich', 'eich@javascript.com', '939570', '2020-05-07', 8),
-> ('4', 'Evan You', 'you@vuejs.org', '982630', '2020-06-11', 7),
-> ('5', 'Rasmus Lerdorf', 'lerdorf@php.net', '937927', '2020-06-03', 8),

-> ('6', 'Guido van Rossum', 'guido@python.org', '968827', '2020-07-18', 19),
-> ('7', 'Adrian Holovaty', 'adrian@django project.com', '570724', '2020-05-07', 5),
-> ('8', 'Simon Willison', 'simon@django project.com', '864615', '2020-04-30', 4),
-> ('9', 'James Gosling', 'james@java.com', '719491', '2020-05-18', 5),
-> ('10', 'Rod Johnson', 'rod@spring.io', '601744', '2020-05-18', 7),
-> ('11', 'Satoshi Nakamoto', 'nakamoto@blockchain.com', '630488', '2020-05-10', 10);
```

```
Query OK, 11 rows affected (0.012 sec)
```

```
Records: 11 Duplicates: 0 Warnings: 0
```

```
MariaDB [Google_22201344]> SELECT *
```

```
-> FROM Developers;
```

member_id	name	email	influence_count	Joining_date	multiplier
1	Taylor Otwell	otwell@laravel.com	739360	2020-06-10	10
2	Ryan Dahl	ryan@nodejs.org	633632	2020-04-22	10
3	Brendan Eich	eich@javascript.com	939570	2020-05-07	8
4	Evan You	you@vuejs.org	982630	2020-06-11	7
5	Rasmus Lerdorf	lerdorf@php.net	937927	2020-06-03	8
6	Guido van Rossum	guido@python.org	968827	2020-07-18	19
7	Adrian Holovaty	adrian@django project.com	570724	2020-05-07	5
8	Simon Willison	simon@django project.com	864615	2020-04-30	4
9	James Gosling	james@java.com	719491	2020-05-18	5
10	Rod Johnson	rod@spring.io	601744	2020-05-18	7
11	Satoshi Nakamoto	nakamoto@blockchain.com	630488	2020-05-10	10

```
11 rows in set (0.000 sec)
```

No 2 Query (as Plain Text)	<pre>SELECT * FROM Developers; ALTER TABLE Developers CHANGE COLUMN influence_count followers int; SELECT * FROM Developers;</pre>

No 2 SS
(of Query & Output
in Shell)

```
MariaDB [Google_22201344]> SELECT *
-> FROM Developers;
```

member_id	name	email	influence_count	Joining_date	multiplier
1	Taylor Otwell	otwell@laravel.com	739360	2020-06-10	10
2	Ryan Dahl	ryan@nodejs.org	633632	2020-04-22	10
3	Brendan Eich	eich@javascript.com	939570	2020-05-07	8
4	Evan You	you@vuejs.org	982630	2020-06-11	7
5	Rasmus Lerdorf	lerdorf@php.net	937927	2020-06-03	8
6	Guido van Rossum	guido@python.org	968827	2020-07-18	19
7	Adrian Holovaty	adrian@django project.com	570724	2020-05-07	5
8	Simon Willison	simon@django project.com	864615	2020-04-30	4
9	James Gosling	james@java.com	719491	2020-05-18	5
10	Rod Johnson	rod@spring.io	601744	2020-05-18	7
11	Satoshi Nakamoto	nakamoto@blockchain.com	630488	2020-05-10	10

11 rows in set (0.001 sec)

```
MariaDB [Google_22201344]> ALTER TABLE Developers
-> CHANGE COLUMN influence_count followers int;
Query OK, 0 rows affected (0.026 sec)
Records: 0 Duplicates: 0 Warnings: 0
```

```
MariaDB [Google_22201344]> SELECT *
-> FROM Developers;
```

member_id	name	email	followers	Joining_date	multiplier
1	Taylor Otwell	otwell@laravel.com	739360	2020-06-10	10
2	Ryan Dahl	ryan@nodejs.org	633632	2020-04-22	10
3	Brendan Eich	eich@javascript.com	939570	2020-05-07	8
4	Evan You	you@vuejs.org	982630	2020-06-11	7
5	Rasmus Lerdorf	lerdorf@php.net	937927	2020-06-03	8
6	Guido van Rossum	guido@python.org	968827	2020-07-18	19
7	Adrian Holovaty	adrian@django project.com	570724	2020-05-07	5
8	Simon Willison	simon@django project.com	864615	2020-04-30	4
9	James Gosling	james@java.com	719491	2020-05-18	5
10	Rod Johnson	rod@spring.io	601744	2020-05-18	7
11	Satoshi Nakamoto	nakamoto@blockchain.com	630488	2020-05-10	10

11 rows in set (0.001 sec)

No 3 Query (as Plain Text)	UPDATE Developers SET followers= followers+10; SELECT * FROM Developers;																																																																								
No 3 SS (of Query & Output in Shell)	<pre>MariaDB [Google_22201344]> UPDATE Developers -> SET followers= followers+10; Query OK, 11 rows affected (0.009 sec) Rows matched: 11 Changed: 11 Warnings: 0 MariaDB [Google_22201344]> SELECT * -> FROM Developers;</pre> <table><tr><th>member_id</th><th>name</th><th>email</th><th>followers</th><th>Joining_date</th><th>multiplier</th></tr><tr><td>1</td><td>Taylor Otwell</td><td>otwell@laravel.com</td><td>739370</td><td>2020-06-10</td><td>10</td></tr><tr><td>2</td><td>Ryan Dahl</td><td>ryan@nodejs.org</td><td>633642</td><td>2020-04-22</td><td>10</td></tr><tr><td>3</td><td>Brendan Eich</td><td>eich@javascript.com</td><td>939580</td><td>2020-05-07</td><td>8</td></tr><tr><td>4</td><td>Evan You</td><td>you@vuejs.org</td><td>982640</td><td>2020-06-11</td><td>7</td></tr><tr><td>5</td><td>Rasmus Lerdorf</td><td>lerdorf@php.net</td><td>937937</td><td>2020-06-03</td><td>8</td></tr><tr><td>6</td><td>Guido van Rossum</td><td>guido@python.org</td><td>968837</td><td>2020-07-18</td><td>19</td></tr><tr><td>7</td><td>Adrian Holovaty</td><td>adrian@django project.com</td><td>570734</td><td>2020-05-07</td><td>5</td></tr><tr><td>8</td><td>Simon Willison</td><td>simon@django project.com</td><td>864625</td><td>2020-04-30</td><td>4</td></tr><tr><td>9</td><td>James Gosling</td><td>james@java.com</td><td>719501</td><td>2020-05-18</td><td>5</td></tr><tr><td>10</td><td>Rod Johnson</td><td>rod@spring.io</td><td>601754</td><td>2020-05-18</td><td>7</td></tr><tr><td>11</td><td>Satoshi Nakamoto</td><td>nakamoto@blockchain.com</td><td>630498</td><td>2020-05-10</td><td>10</td></tr></table> <pre>11 rows in set (0.000 sec)</pre>	member_id	name	email	followers	Joining_date	multiplier	1	Taylor Otwell	otwell@laravel.com	739370	2020-06-10	10	2	Ryan Dahl	ryan@nodejs.org	633642	2020-04-22	10	3	Brendan Eich	eich@javascript.com	939580	2020-05-07	8	4	Evan You	you@vuejs.org	982640	2020-06-11	7	5	Rasmus Lerdorf	lerdorf@php.net	937937	2020-06-03	8	6	Guido van Rossum	guido@python.org	968837	2020-07-18	19	7	Adrian Holovaty	adrian@django project.com	570734	2020-05-07	5	8	Simon Willison	simon@django project.com	864625	2020-04-30	4	9	James Gosling	james@java.com	719501	2020-05-18	5	10	Rod Johnson	rod@spring.io	601754	2020-05-18	7	11	Satoshi Nakamoto	nakamoto@blockchain.com	630498	2020-05-10	10
member_id	name	email	followers	Joining_date	multiplier																																																																				
1	Taylor Otwell	otwell@laravel.com	739370	2020-06-10	10																																																																				
2	Ryan Dahl	ryan@nodejs.org	633642	2020-04-22	10																																																																				
3	Brendan Eich	eich@javascript.com	939580	2020-05-07	8																																																																				
4	Evan You	you@vuejs.org	982640	2020-06-11	7																																																																				
5	Rasmus Lerdorf	lerdorf@php.net	937937	2020-06-03	8																																																																				
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No 4 Query	SELECT name,																																																																								

(as Plain Text)

((followers*100/1000000)*(multiplier*100/20))/100 AS
Efficiency
FROM Developers;

No 4 SS
(of Query & Output
in Shell)

```
MariaDB [Google_22201344]> SELECT name,  
-> ((followers*100/1000000)*(multiplier*100/20))/100 AS  
-> Efficiency  
-> FROM Developers;
```

name	Efficiency
Taylor Otwell	36.968500000000
Ryan Dahl	31.682100000000
Brendan Eich	37.583200000000
Evan You	34.392400000000
Rasmus Lerdorf	37.517480000000
Guido van Rossum	92.039515000000
Adrian Holovaty	14.268350000000
Simon Willison	17.292500000000
James Gosling	17.987525000000
Rod Johnson	21.061390000000
Satoshi Nakamoto	31.524900000000

11 rows in set (0.001 sec)

No 5 Query (as Plain Text)	SELECT UPPER(name) AS name, Joining_date FROM Developers ORDER BY Joining_date DESC;
No 5 SS (of Query & Output in Shell)	<pre> MariaDB [Google_22201344]> SELECT UPPER(name) AS name, Joining_date -> FROM Developers -> ORDER BY Joining_date DESC; +-----+-----+ name Joining_date +-----+-----+ GUIDO VAN ROSSUM 2020-07-18 EVAN YOU 2020-06-11 TAYLOR OTWELL 2020-06-10 RASMUS LERDORF 2020-06-03 JAMES GOSLING 2020-05-18 ROD JOHNSON 2020-05-18 SATOSHI NAKAMOTO 2020-05-10 BRENDAN EICH 2020-05-07 ADRIAN HOLOVATY 2020-05-07 SIMON WILLISON 2020-04-30 RYAN DAHL 2020-04-22 +-----+-----+ 11 rows in set (0.001 sec) </pre>
No 6 Query (as Plain Text)	SELECT member_id, name, email, followers

	FROM Developers WHERE email like '%.com' OR email like '%.net';																																
No 6 SS (of Query & Output in Shell)	<pre>MariaDB [Google_22201344]> SELECT -> member_id, name, email, followers -> FROM Developers -> WHERE email like '%.com' OR email like '%.net';</pre> <table><tr><th>member_id</th><th>name</th><th>email</th><th>followers</th></tr><tr><td>1</td><td>Taylor Otwell</td><td>otwell@laravel.com</td><td>739370</td></tr><tr><td>3</td><td>Brendan Eich</td><td>eich@javascript.com</td><td>939580</td></tr><tr><td>5</td><td>Rasmus Lerdorf</td><td>lerdorf@php.net</td><td>937937</td></tr><tr><td>7</td><td>Adrian Holovaty</td><td>adrian@django project.com</td><td>570734</td></tr><tr><td>8</td><td>Simon Willison</td><td>simon@django project.com</td><td>864625</td></tr><tr><td>9</td><td>James Gosling</td><td>james@java.com</td><td>719501</td></tr><tr><td>11</td><td>Satoshi Nakamoto</td><td>nakamoto@blockchain.com</td><td>630498</td></tr></table> <pre>7 rows in set (0.001 sec)</pre>	member_id	name	email	followers	1	Taylor Otwell	otwell@laravel.com	739370	3	Brendan Eich	eich@javascript.com	939580	5	Rasmus Lerdorf	lerdorf@php.net	937937	7	Adrian Holovaty	adrian@django project.com	570734	8	Simon Willison	simon@django project.com	864625	9	James Gosling	james@java.com	719501	11	Satoshi Nakamoto	nakamoto@blockchain.com	630498
member_id	name	email	followers																														
1	Taylor Otwell	otwell@laravel.com	739370																														
3	Brendan Eich	eich@javascript.com	939580																														
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7	Adrian Holovaty	adrian@django project.com	570734																														
8	Simon Willison	simon@django project.com	864625																														
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No 7 Query (as Plain Text)																																	
No 7 SS (of Query & Output in Shell)																																	